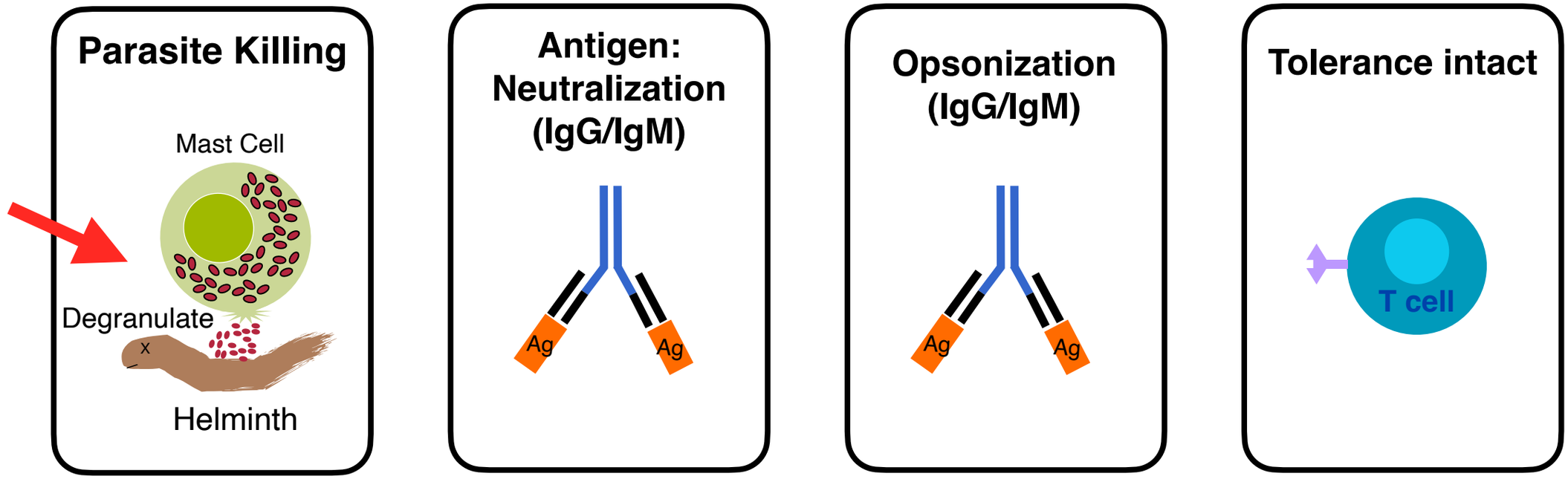
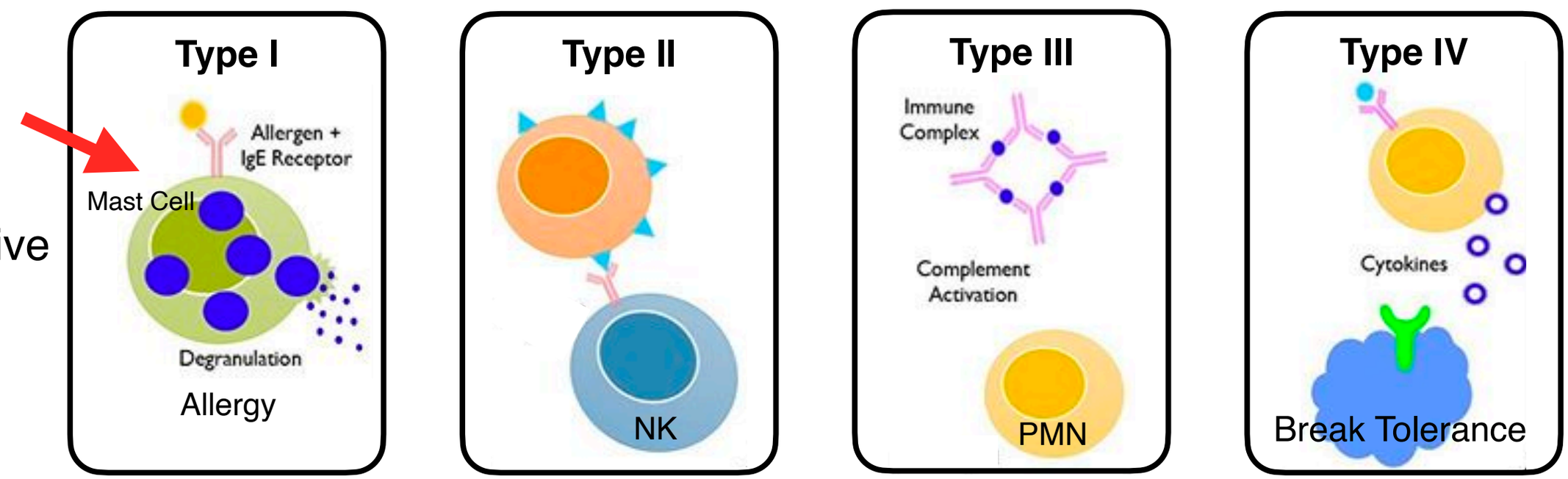
Hypersensitivity Basics

Normal



Abnormal Hypersensitive Responses



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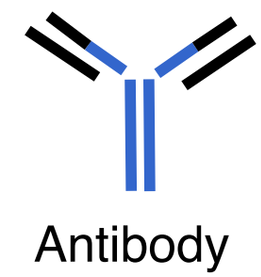
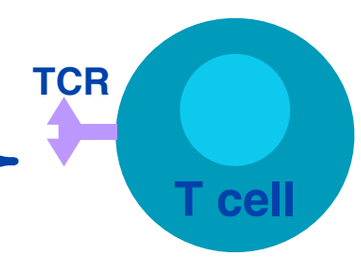
Session Objectives

- Describe the immunological mechanisms underlying:
  - Type I (Immediate) Hypersensitivity reactions
  - Type II (Antibody-mediated) Hypersensitivity reactions
  - Type III (Immune-complex mediated) Hypersensitivity reactions
  - Type IV (Cell-mediated) Hypersensitivity reactions

The A, B, C, and Ds of Hypersensitivity

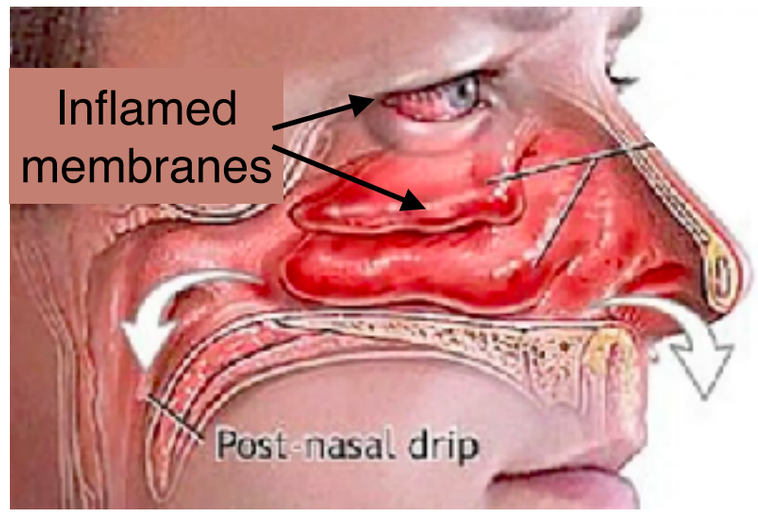
Type I	Immediate Hypersensitivity **A**ntibody-**A**sthma-**A**naphylaxis
Type II	Antibody **B**inding to host cells/tissues (cytotoxic)
Type III	Antibody/Antigens forming Immune **C**omplexes
Type IV	**D**elayed Type T cell-mediated Hypersensitivity



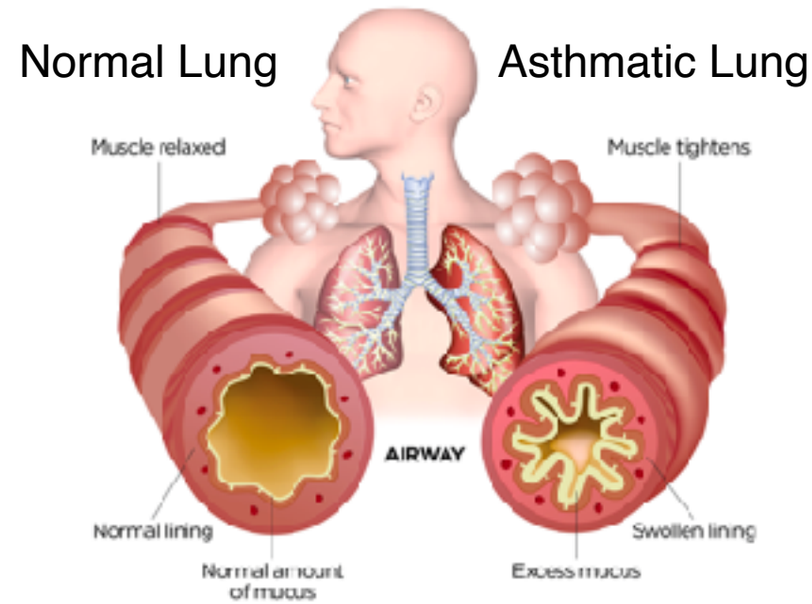


Immediate Hypersensitivity (Type I)

Hives



Allergic Rhinitis and Conjunctivitis



Asthma

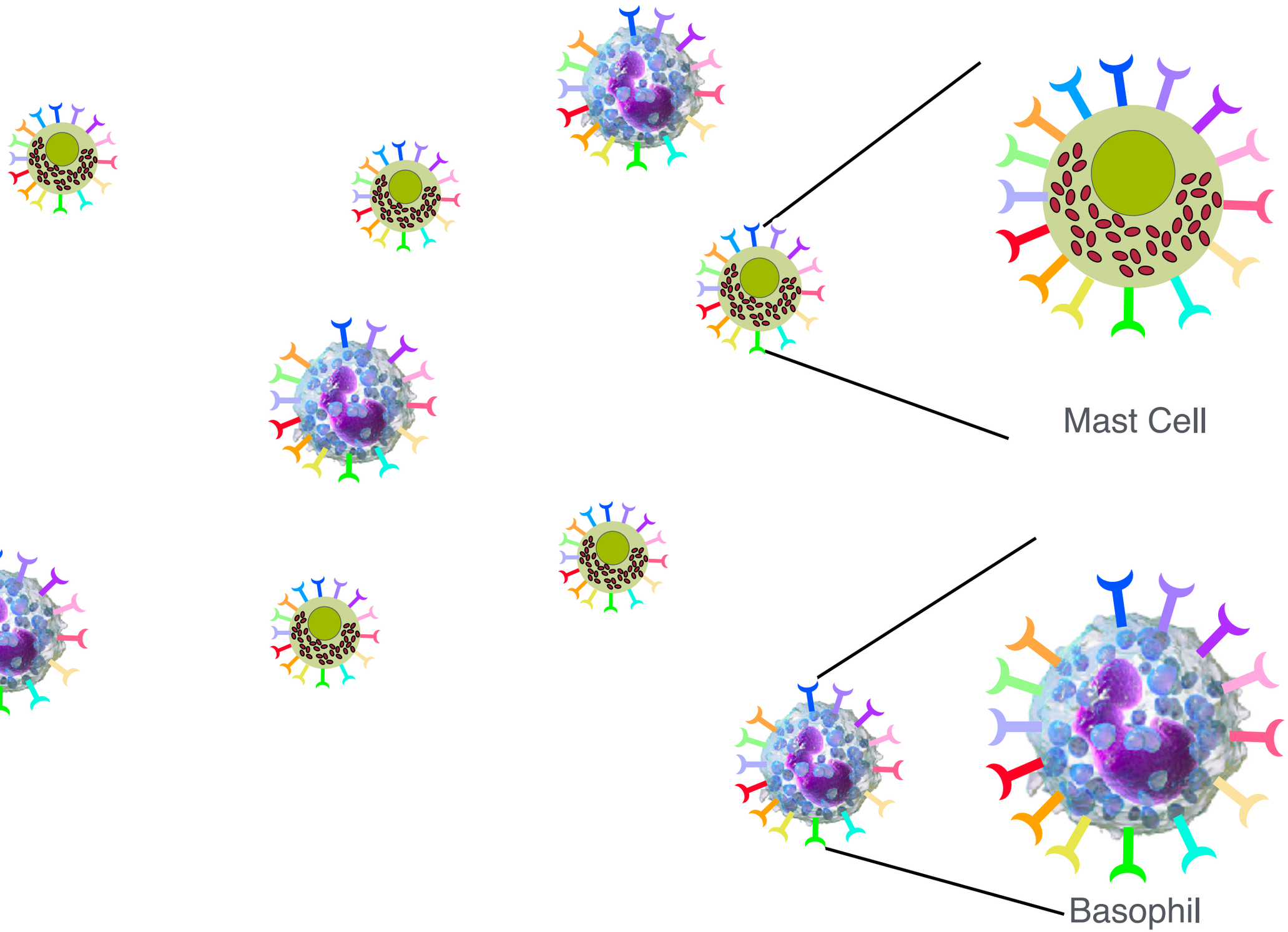
Food-induced diarrhea and vomiting

Anaphylaxis

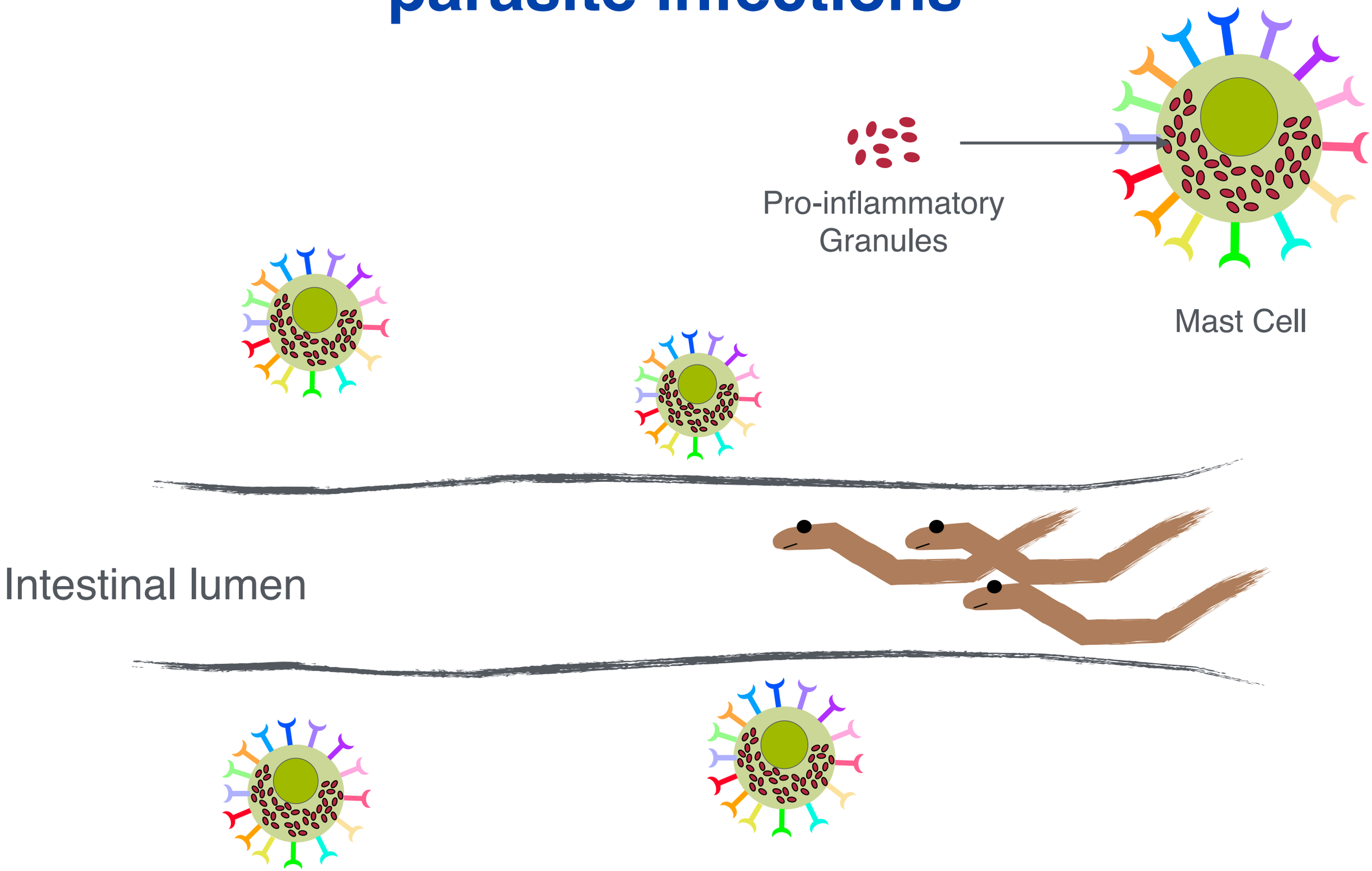
Systemic

- severe bronchoconstriction
- shock

Mast Cells and Basophils



Mast Cells are important for controlling parasite infections



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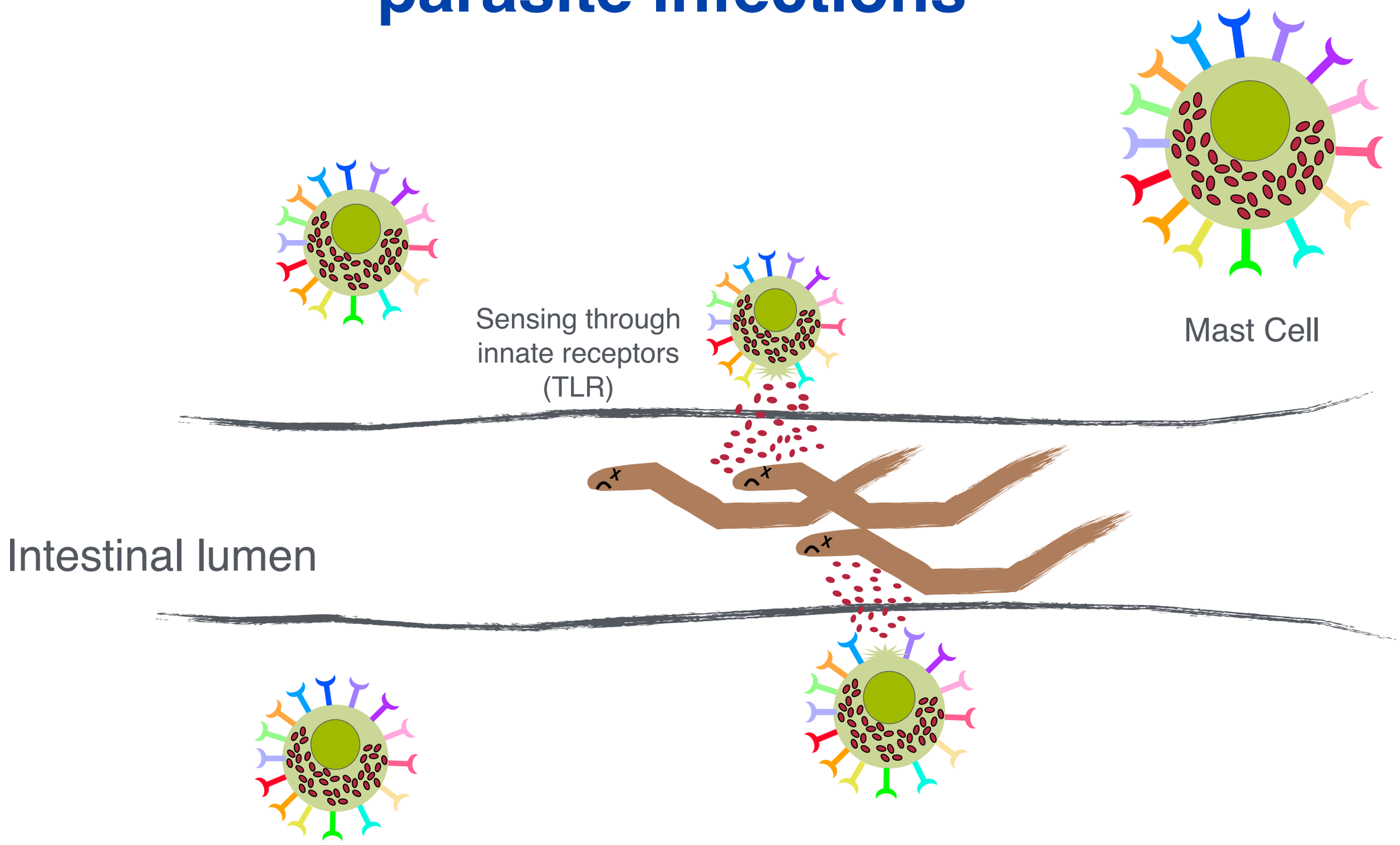
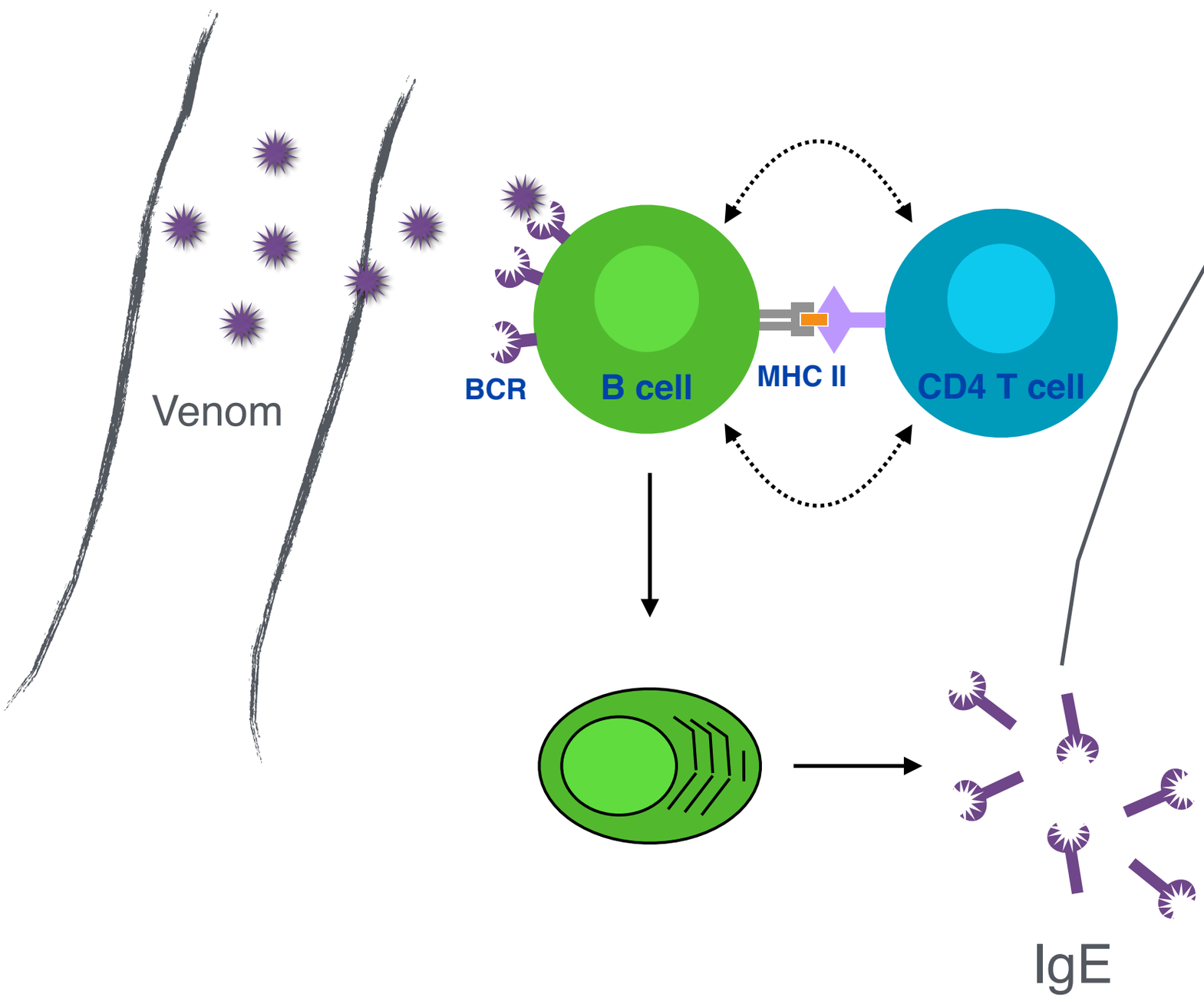
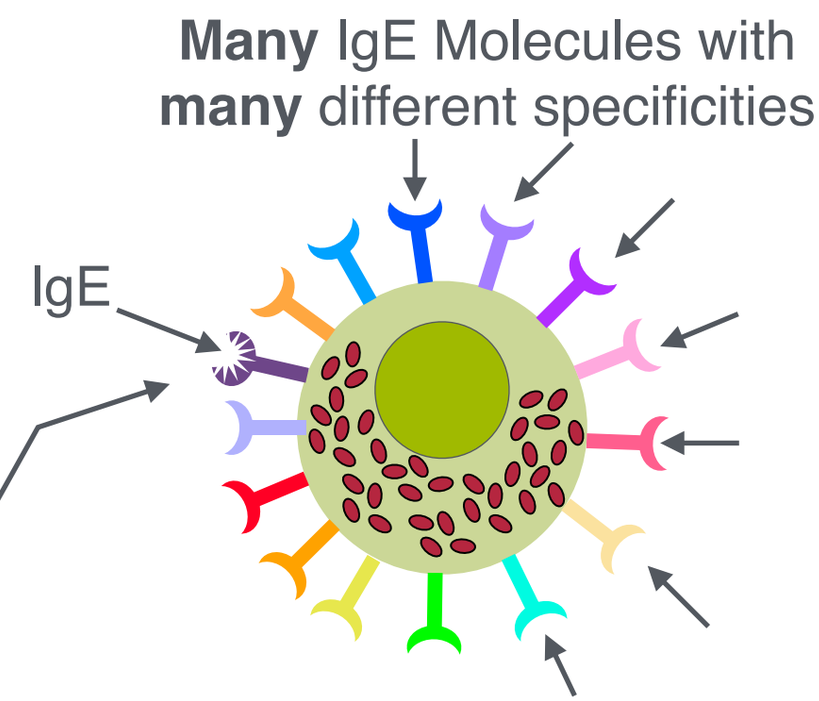


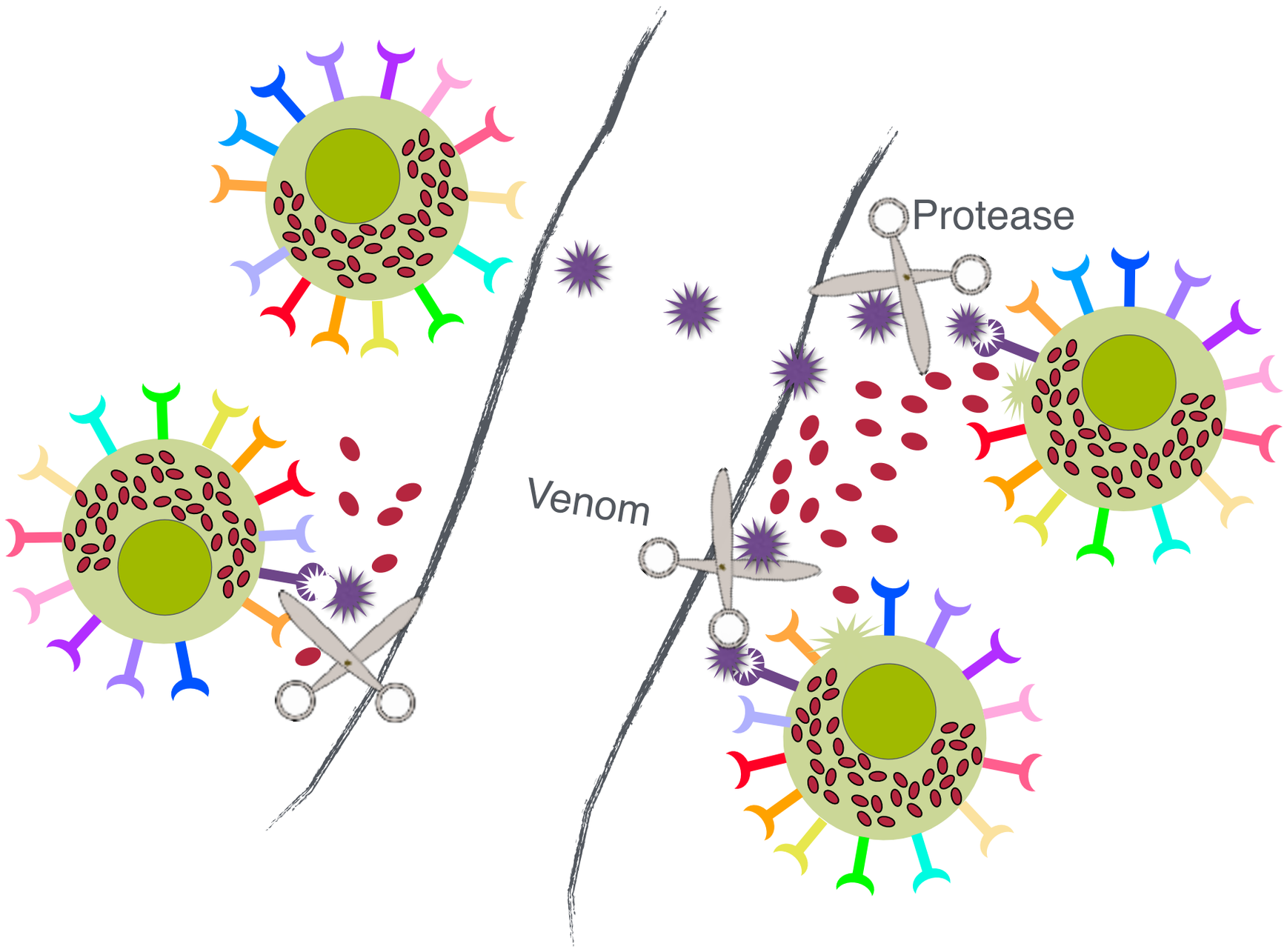
Mast Cells are important for responding to certain antigens like venom



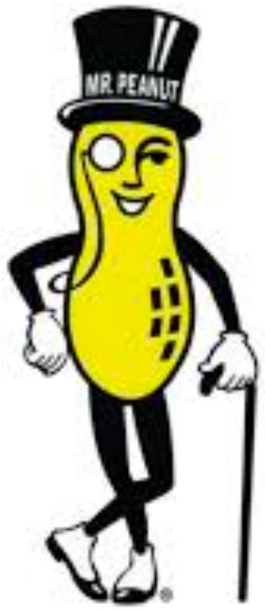


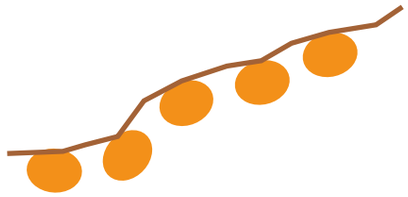
Mast Cells also express a high affinity Fcε Receptor that binds IgE antibody

Mast Cells are important for responding to certain antigens like venom

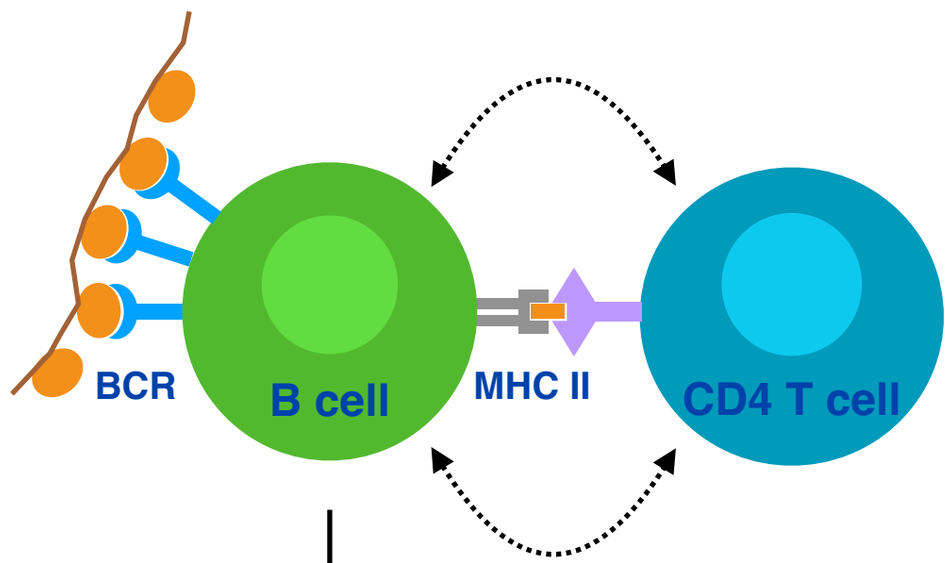


Mast Cells are also important in Type I Hypersensitivity Reactions

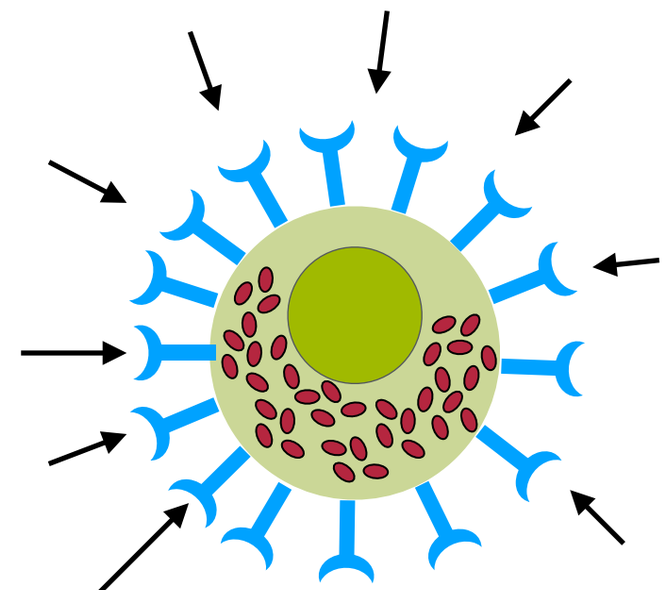


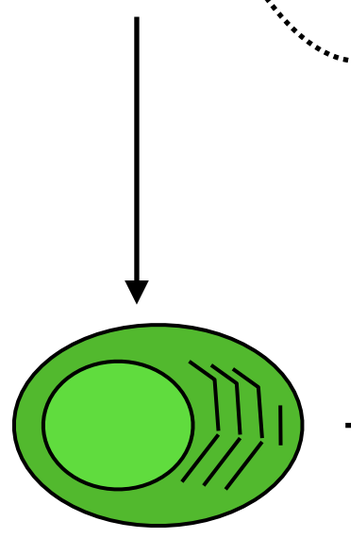


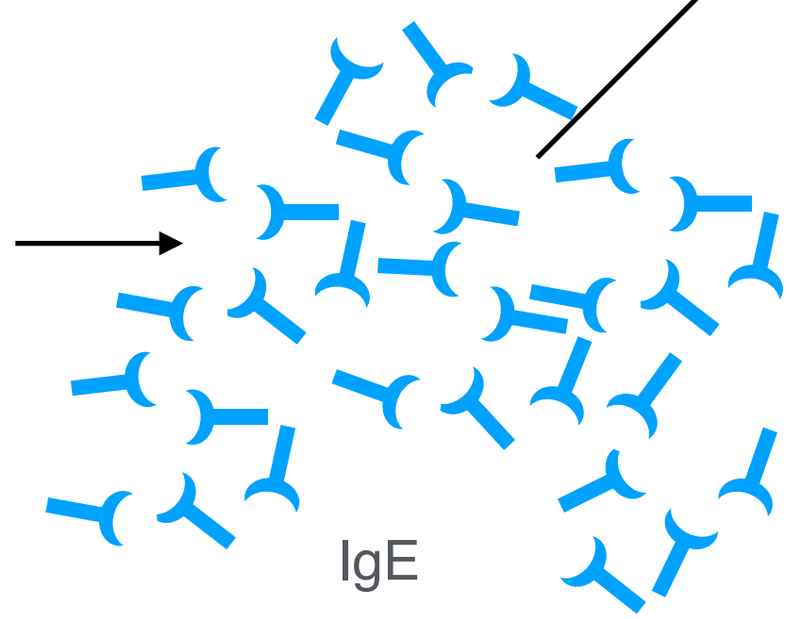
Peanut allergen



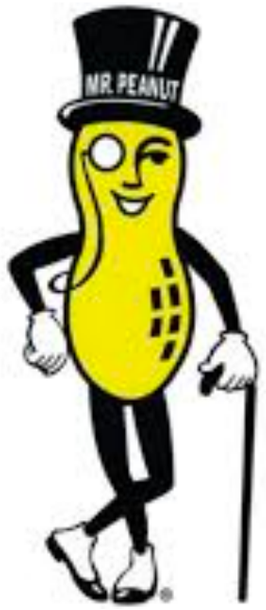
Many IgE Molecules with **the same** specificity

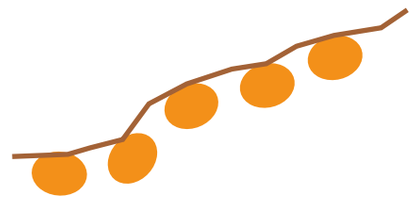




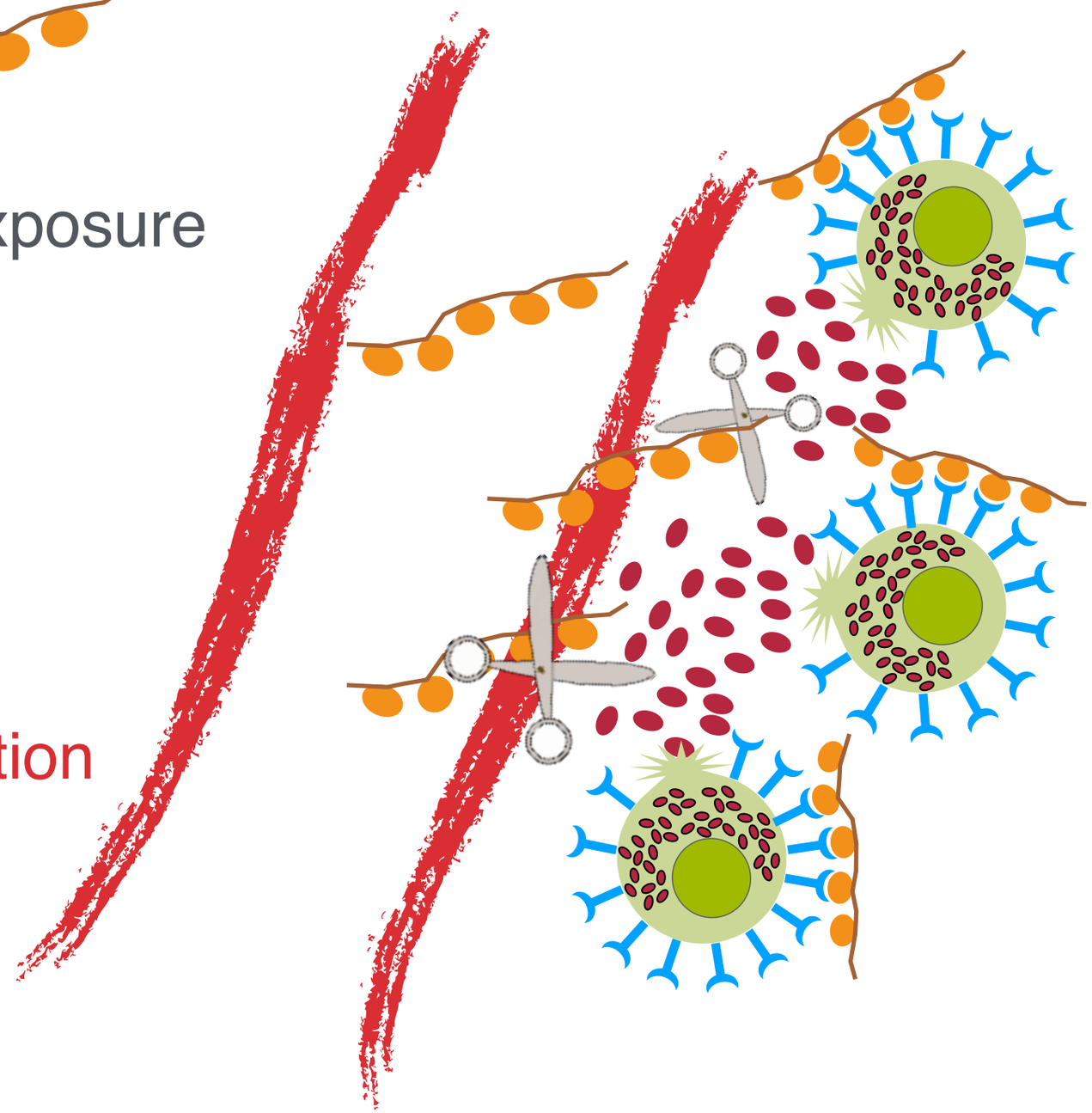


Mast Cells are also important in Type I Hypersensitivity Reactions





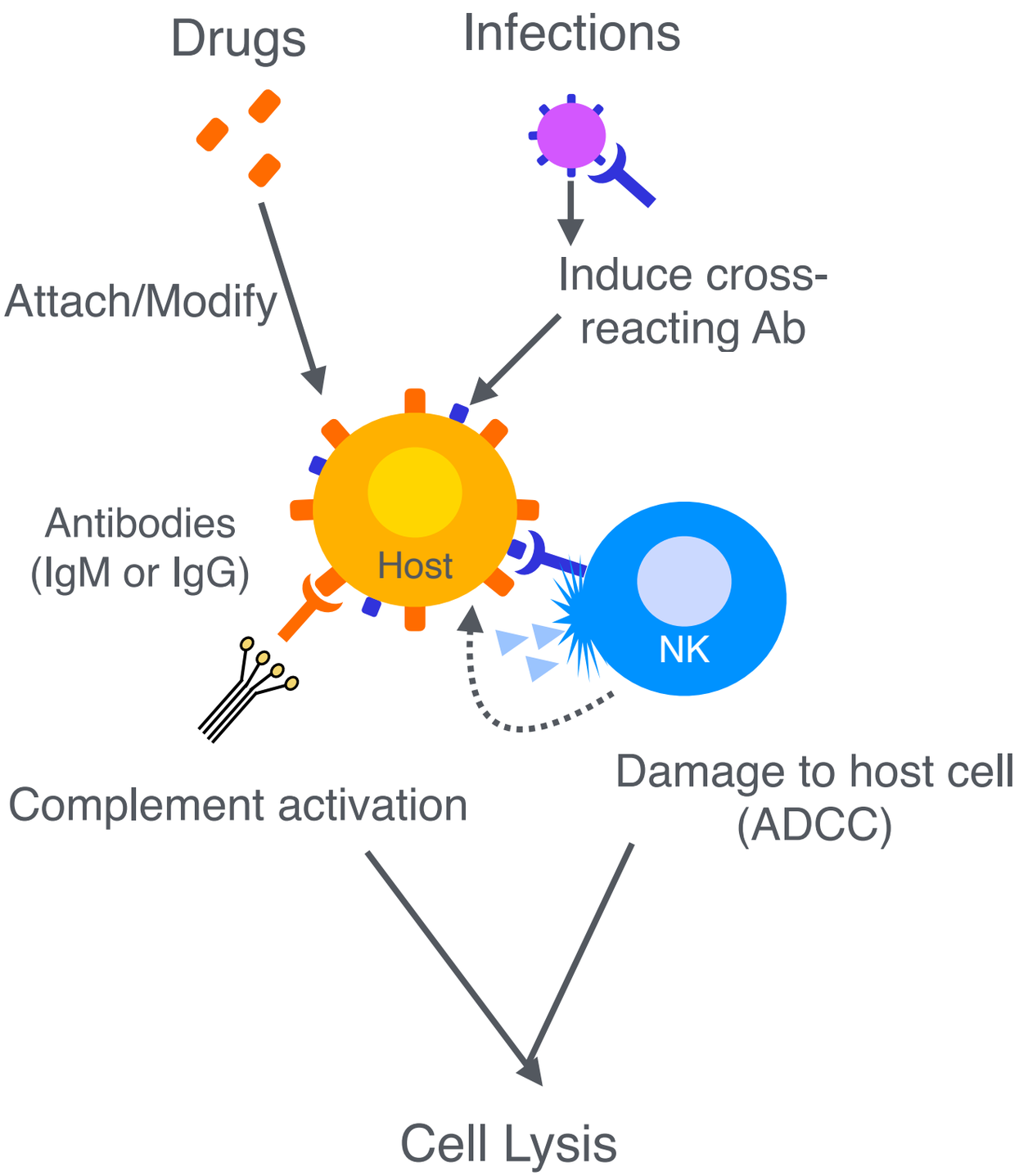
Repeat Exposure



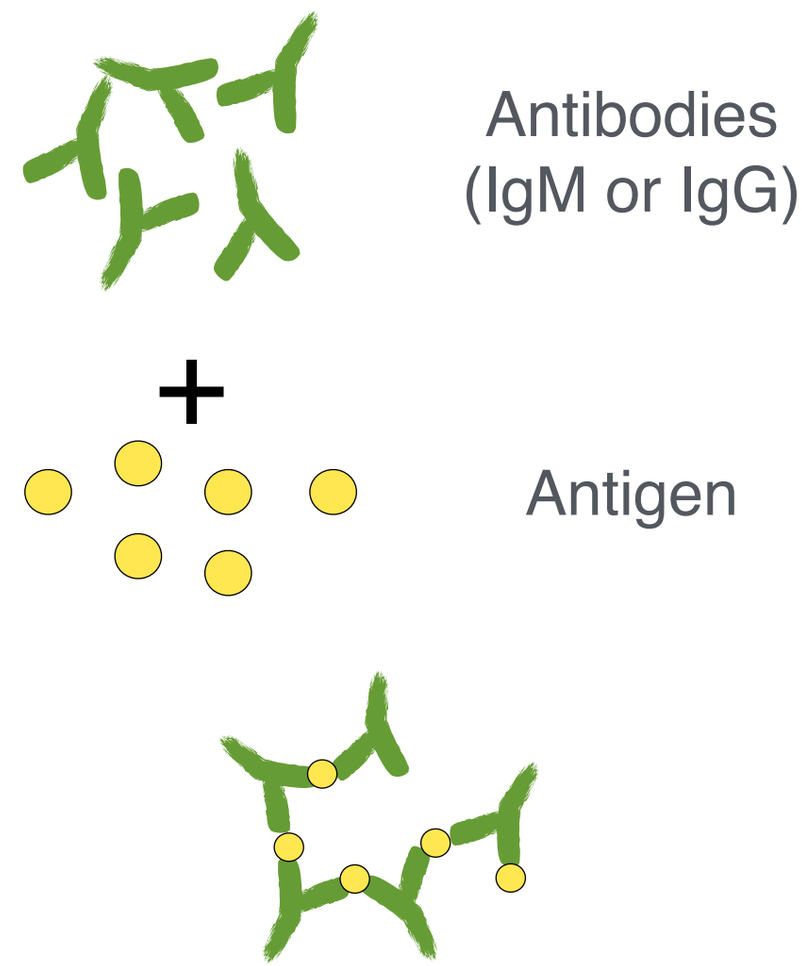
Vasodilation

Large scale release of pro inflammatory mediators.
Can lead to anaphylaxis within minutes of exposure

Type II Hypersensitivity: Cross-reactive Ab



Type III Hypersensitivity: Immune Complexes



Immune Complexes

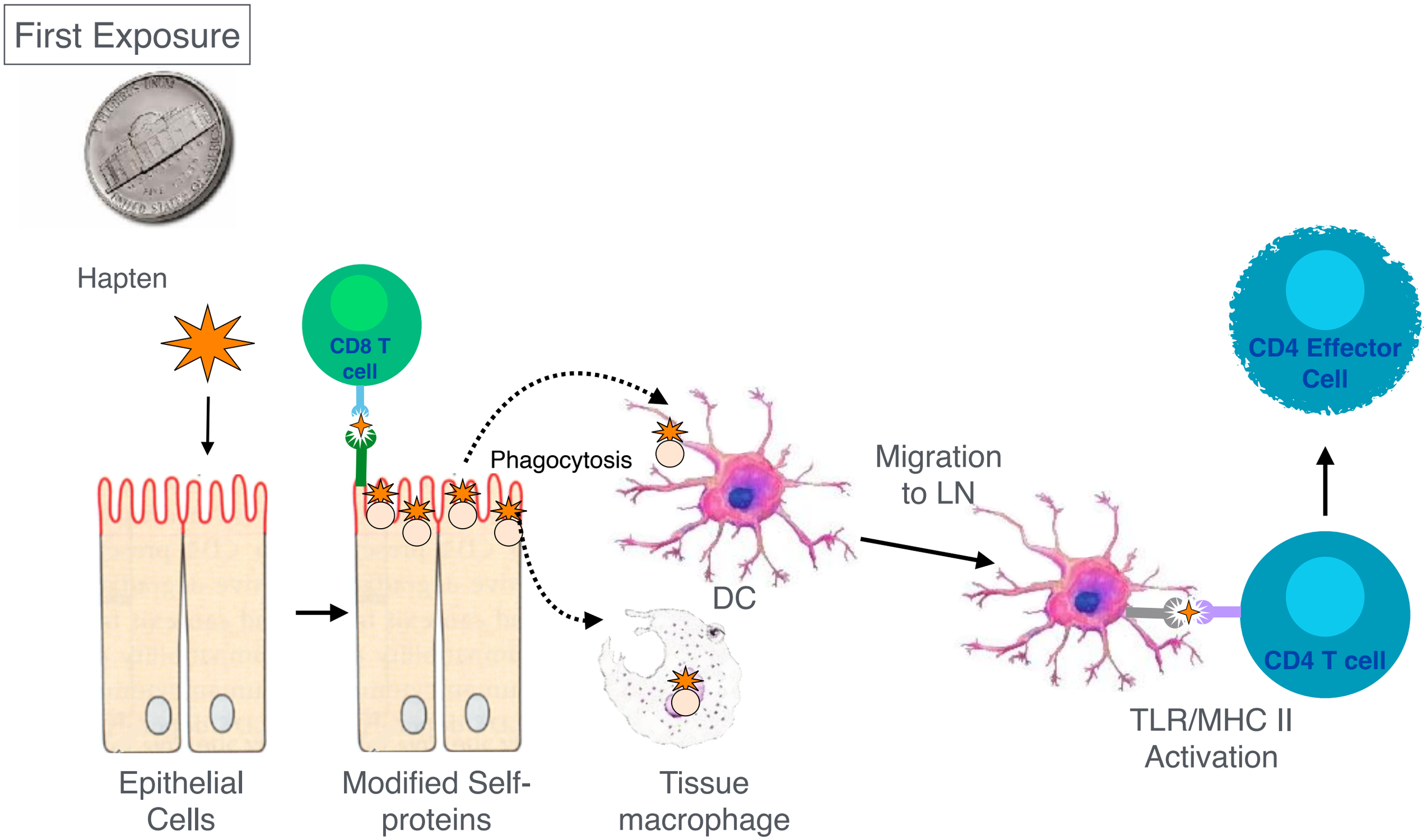
↓

Complement activation

↓

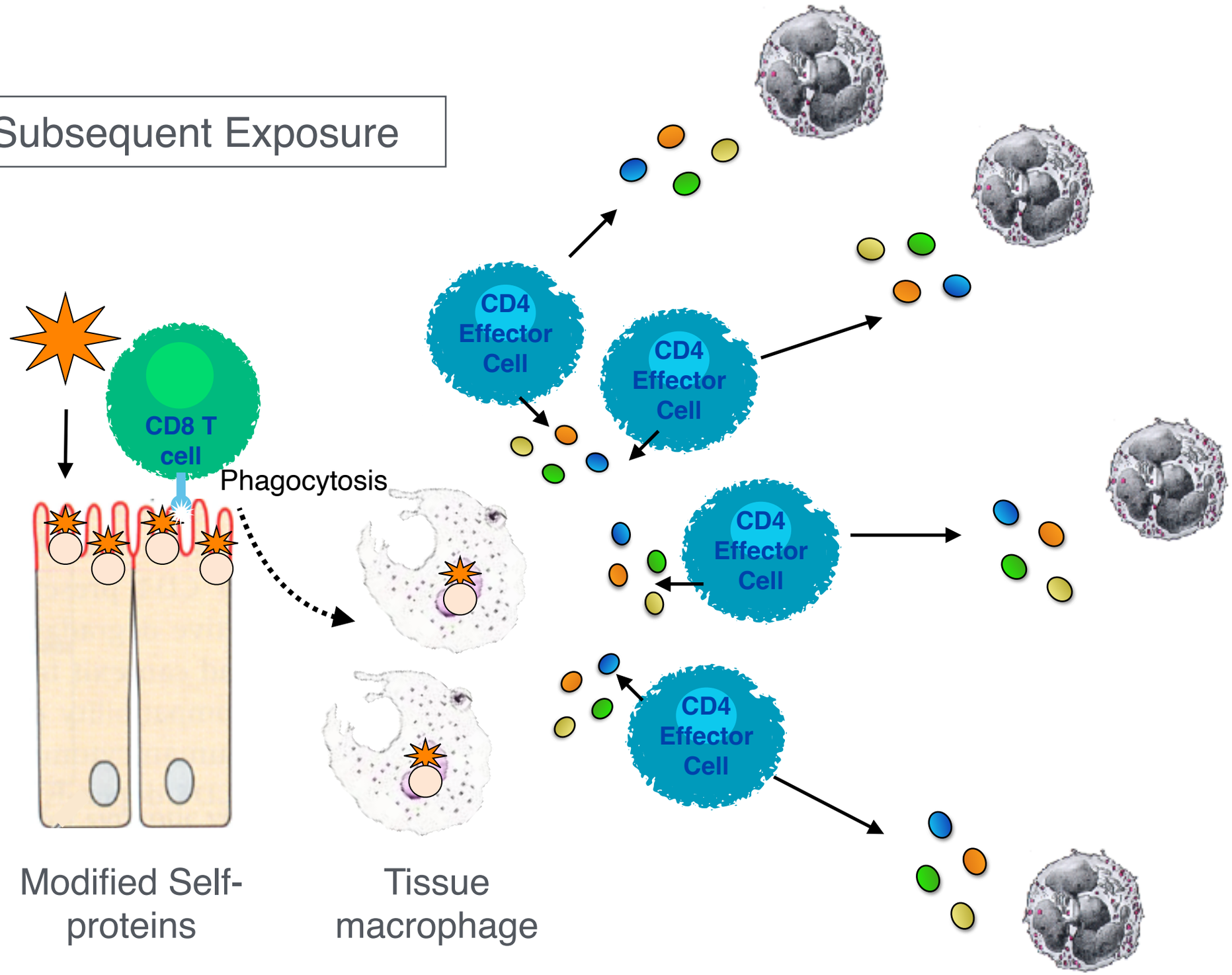
Cell Lysis

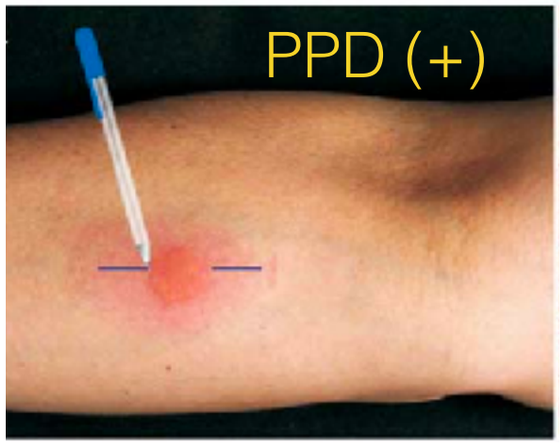
Type IV Hypersensitivity: Delayed T cell-mediated

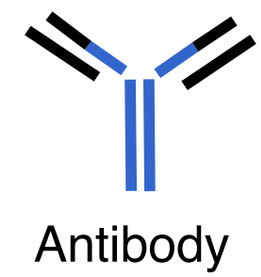
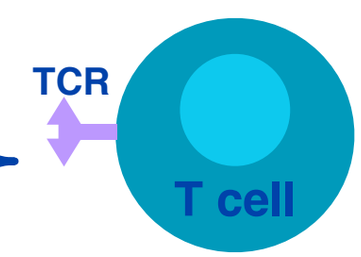


Type IV Hypersensitivity: Delayed T cell-mediated

Subsequent Exposure





The A, B, C, and Ds of Hypersensitivity

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